LIVING OUT OF SYNC: NEUROANATOMICAL CORRELATES OF TIME DISORIENTATION IN PATIENTS WITH FOCAL BRAIN LESIONS

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Introduction

Orientation describes a person's awareness in three dimensions: person, place, and time. Assessing a patient's orientation is a cornerstone of mental status exams for suspected neurological or psychiatric abnormalities. Orientation to time is the most fragile flavor of orientation, being the first to deteriorate following brain injury and the last to recover.

Despite the frequency with which time orientation is used to assess patients, little is known about how the brain supports this function. Multiple neurological and psychiatric disease states can lead to time disorientation, which suggests several regions contribute to normal functioning. This study sought to elucidate the **neuroanatomical correlates of time orientation** in human patients with focal brain lesions from both a regional and network perspective.

We hypothesized that damage to the default mode network, particularly the **medial temporal lobes** and **orbitofrontal cortex**, would be associated with chronic (persisting 3 or more months post-onset) time disorientation as measured by the Benton Temporal Orientation Scale (BTOS).

Methods

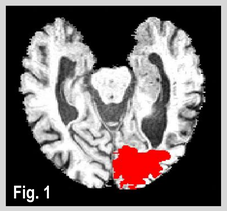


Figure 1.Trace patient's focal lesion in native brain space

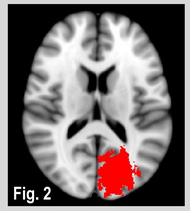


Figure 2.Warp all lesions to MNI152 space for group-level comparison

Fig. 3 $$\frac{lesion\ \&\ deficit}{all\ patients\ with\ deficit} - \frac{lesion\ \&\ no\ deficits}{all\ patients\ with\ no\ deficits}$$

Figure 3. Equation used in proportional subtraction analysis



Figure 4. Separate grey matter (orange) and white matter (blue) findings

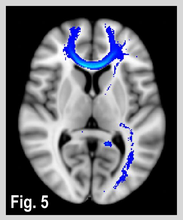


Figure 5. Seed fiber tractography analysis conducted using LEAD-DBS with 3mm regions of interest (ROIs) around peak voxels in white matter regions

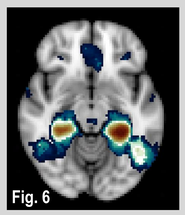


Figure 6. Seed functional connectivity magnetic resonance imaging (fcMRI) with grey matter clusters identified using FSL cluster

Neuropsychological Data

Table 1 **Demographics**

Characteristic	All Patients (N=450)	Impaired (n=20)	Unimpaired (n=430)
Age in years [mean (std)]	52.66 (14.35)	61.99 (15.46)	52.22 (14.17)
Education in years [mean (std)]	13.34 (2.85)	12.35 (4.57)	13.39 (2.74)
Gender (Female, Male)	209 / 241	10 / 10	199 / 231
Handedness (Right, Left, Mixed)	395 / 41 / 14	15 / 4 / 1	380 / 37 / 13
Race (American Indian, African American, Caucasian, Unknown)	1 / 7 / 441 / 1	0 / 0 / 20 / 0	1 / 7 / 421 / 1
Ethnicity (Hispanic, non-Hispanic, unknown)	2 / 447 / 1	0 / 0 / 20	2 / 427 / 1
Lesion Volume in cubic mm [mean (std)]	49347 (56366)	69767 (49622)	48388 (56533)
Lesion Laterality (Right, Left, Bilateral)	161 / 198 / 91	7 / 3 / 10	154 / 195 / 81
Etiology (stroke, SAH, resection, head trauma, encephalitis, multiple etiologies)	304 / 24 / 87 / 14 / 12 / 9	9 / 2 / 2 / 2 / 5 / 0	295 / 22 / 85 / 12 / 7 / 9

Table 1. Impairment on the BTOS is defined as a score of -10 or more points based on verbal response to the following five questions:

1. What month is it? (-5 per month, max error -30)
2. What day of the month is it? (-1 per day, max error -15)
3. What year is it? (-10 per year, max error -60)
4. What day of the week is it? (-1 per day, max error -3)
5. What time of day is it? (-1 per half hour, max error -5)

Overlap Maps

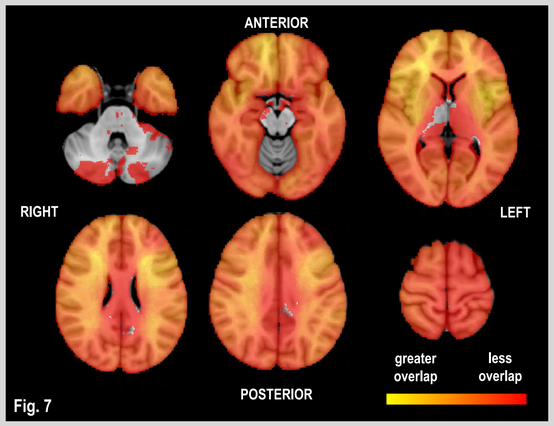


Figure 7. Lesion masks of all 450 patients were overlapped to reveal coverage in all cortical areas.

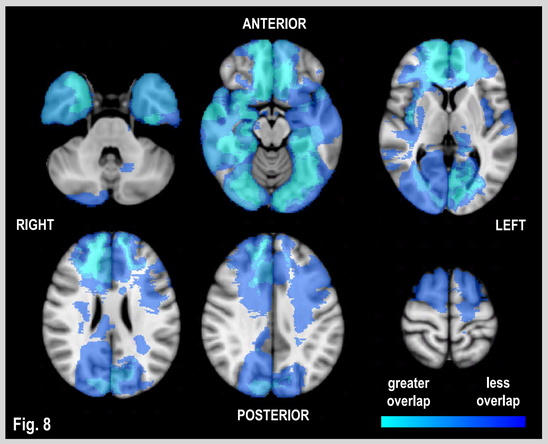


Figure 8. Lesion masks of all impaired patients were overlapped.

Proportional Subtraction

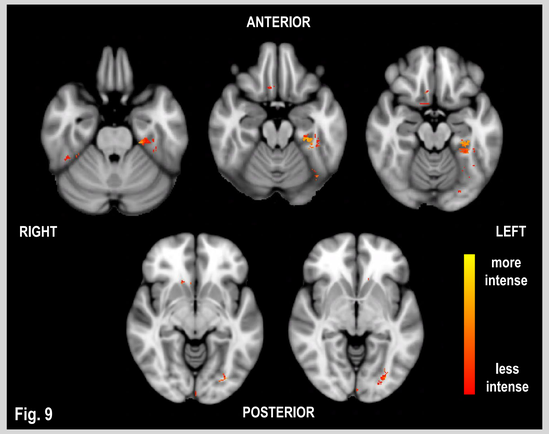


Figure 9. The proportional subtraction map indicates that lesions to the **right straight gyrus** and **left parahippocampal gyrus** are associated with time disorientation.

Fiber Tractography

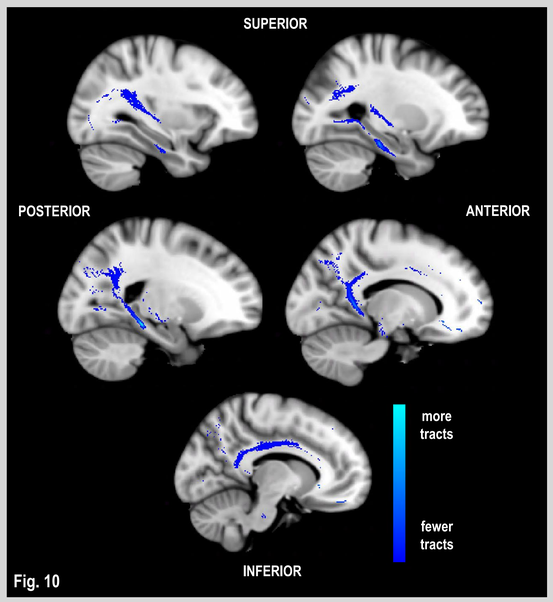


Figure 10. This figure shows the right hemisphere. We created 3mm spherical ROIs around peak white matter voxels from the proportional subtraction analysis to seed a fiber tractography analysis. ROIs were placed in white matter deep to the straight gyrus and white matter surrounding the parahippocampal gyrus. The fiber tractography analysis used diffusion magnetic resonance imaging from 842 healthy adults to determine which white matter bundles are affected by lesions associated with time disorientation.

Functional Lesion Network Mapping

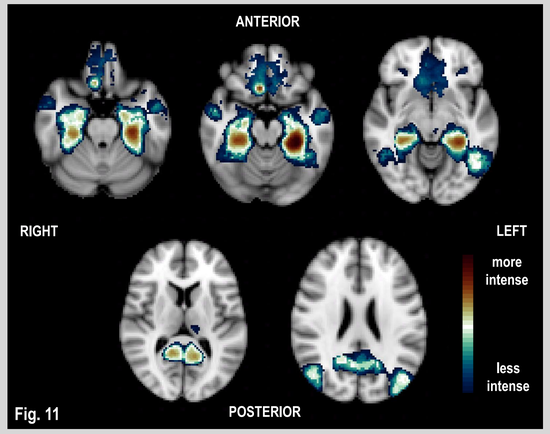


Figure 11. Functional lesion network mapping (Boes et al., 2015) uses functional connectivity magnetic resonance imaging data from 98 healthy adults to determine which functional networks are associated with time disorientation. Peak voxels in the ventromedial prefrontal cortex and parahippocampal gyrus were identified using FSL cluster and used to seed this analysis. Based on the Yeo 7 parcellation scheme (Yeo et al., 2011), lesions associated with time disorientation may impact the limbic network, the dorsal attention network, and the default mode network.

Conclusions

This study investigated neuroanatomical correlates of time disorientation in lesion patients. We found **that damage to the right straight gyrus and left parahippocampal gyrus is associated with impairment on the BTOS**. Damage to white matter adjacent to these regions may have widespread effects on other structurally connected regions. Further, the limbic, dorsal attention, and default mode networks may be affected by these lesions. Together, our results suggest that time orientation is a network-level cognitive ability that requires multiple structurally and functionally connected brain regions for normal performance. Future studies could examine the impact of lesions to these regions on other aspects of temporal cognition, like interval timing.

Acknowledgements & References

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